

Error analysis

Group: Arabic 6

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In this paper we will be examining the kinds of mistakes our ARABIC to English emotion classification model is making. The model is a transformer-based architecture model and it has been trained to recognize seven emotions: happiness, sadness, anger, surprise, fear, disgust, and neutral. This analysis looks at why the model isn't performing and states recommendations on how to deal with the issue at hand.

1. Imbalanced Distribution of Emotion Labels

One of the problems we noticed was that some emotions were mentioned more than others in the dataset. Emotions like "neutral" and "happiness" were common, while emotions like "disgust" and "fear" were used much less frequently. This unbalanced distribution caused the model to prefer predicting the more frequent ones.

Emotion	Amount in Dataset	General Accuracy
Neutral	Very High	Decent
Happiness	High	Good
Sadness	Medium	Lower
Surprise	Medium	Moderate
Anger	Low	Weak
Fear	Very Low	Poor
Disgust	Very Low	Poor

When we looked at the confusion matrix, we saw that rare emotions often got predicted as more common ones, especially neutral or sadness.

2. Sentence Length and Emotion Clarity

The model struggled the most with very short inputs. Sentences under three words were not giving enough contextual information for the model to guess the emotion properly. For example, if a sentence just said “sad,” it was sometimes labeled as neutral.

On the other hand, longer inputs gave more clues and usually helped the model guess more accurately. It’s clear that the amount of contextual information available plays a huge role in model performance.

3. Similar Emotions Get Confused

Some emotions share similar tone or vocabulary, especially when translated from Arabic to English. This led to confusion between certain pairs:

- Sadness and fear were mixed up often
- Anger was sometimes marked as disgust
- Surprise was confused with happiness when the expression was light

Examples:

- Actual label: Fear → Predicted: Sadness
- Actual label: Anger → Predicted: Disgust
- Actual label: Surprise → Predicted: Happiness

In these cases, the problem is probably due to how the emotional tone is translated and interpreted by the model.

4. Translation Problems

Because the model works on English translations of Arabic sentences, translation quality makes a big difference. Arabic slang don’t always come across correctly in English. That can lead to translations that are either too vague or don’t carry the same emotional strength.

For example:

Arabic: "حرقت أعصابي" → English: "My nerves are fried" Label: Anger → Prediction: Sadness

This phrase in Arabic clearly shows frustration, but the English version sounds more passive. The model chose sadness instead of anger, possibly because of this.

5. Confusion Matrix Observations

From the confusion matrix, a few trends stood out:

- The model often chose “neutral” when unsure
- Happiness had decent accuracy but missed some true cases

- Emotions like fear and disgust were rarely predicted correctly

This tells us the model avoids risk and picks safe options like neutral when there isn't a strong signal.

How We Can Improve the Model

1. Fix the Imbalance

- Use more training samples for rare emotions
- Try oversampling or generating new examples using back-translation
- Use weighted loss to give more importance to less common emotions

2. Handle Short Sentences Better

- Filter out extremely short inputs
- Add context manually or through augmentation

3. Better Translation

- Use higher-quality translators or emotion-aware models
- Or skip translation completely and work directly with Arabic using models like AraBERT

Conclusion

The model does okay when working with popular emotions and longer sentences. But it struggles with emotions that don't show up much or that have subtle differences.

Translation adds another layer of complexity. If we can train on more balanced and expressive data and maybe use a better approach to translation or even avoid it, we believe this model can get a lot better in classifying emotions from Arabic text.